

Design Details and Characterization of All Digital Closed-Loop Interferometric Fiber Optic Gyroscope with Superluminescent Light Emitting Diode

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This paper describes the design details and the characterization of an all digital closed-loop (ADCL) interferometric fiber optic gyroscope (IFOG) Prototype with sine wave biasing modulation and SLED emitting at 1288.9 nm, constructed in TUBITAK UME. Electronic feedback section of ADCL-IFOG is composed of 16 bit AD7714YN ADC, 14 bit-LTC1667CG DAC, fully controlled and synchronized by 89C51 microcontroller. The averaged sensitivity of the ADCL-IFOG prototype in unit of error voltage applied to the phase modulator was calculated as 132.65 $\mu\text{V}/(^{\circ}/\text{h})$ which equals to a scale factor of 7.53888 $(^{\circ}/\text{h})/\text{mV}$ with a standard deviation of 0.85% for a range of 1–15270 $(^{\circ}/\text{h})$ rotation rate, corresponding to a range of Sagnac phase shifts varying from 0.00115 to 17.57448 $(^{\circ})$. The maximum peak to peak noise and the bias stability of the prototype were determined as 4.97 and 1.48 $(^{\circ}/\text{h})$ at 23.0 $^{\circ}\text{C}$, respectively. The variation of the noise and bias stability on a temperature range of 23.0–42.0 $^{\circ}\text{C}$ was observed.

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1. Introduction

Gyroscopes are sensor elements, which convert rotation rate into voltage and they are classified into mechanical gyroscopes and optical gyroscopes. Micro-electromechanical systems (MEMS) and micromachined-optoelectronic mechanical systems (MOEMS) gyroscopes constitute the mechanical style gyroscopes and all the mechanical gyroscopes have vibrating elements to detect Coriolis Acceleration, which yields rotation information.^{1,2)} Optical gyroscopes consist of ring laser and fiber optic gyroscopes, which are widely used in navigation applications and their physical principle to sense rotation rate is based on the Sagnac phase shift, which was discovered in 1913 by Sagnac.

Although the first ring interferometer experiment to demonstrate Sagnac phase shift for electromagnetic waves propagating in a rotating media was performed by Harress who encountered with many problems in describing the results of the experiment,³⁾ the first successful attempt to measure the rotation rate was carried out by Michelson and Gale in 1925 with a gigantic Michelson interferometer configuration. With development of optical fiber manufacturing technology, a fiber ring interferometer with multiple optical paths was first constructed by Vali and Shorthill in 1976 to show Sagnac phase shift and this interferometer configuration is known as a pioneering step in the applications of optical fiber gyroscope.³⁾

Sagnac phase shift ϕ_R , which is a function of the rotation rate (angular rate) of an interferometer configuration with respect to an inertial frame, is only induced by rotation, not linear acceleration. Sagnac phase shift ϕ_R for a ring interferometer having N turn optical path is given in the following;

$$\phi_R = \frac{8\pi NA \cdot \Omega}{\lambda_0 c_0}, \quad (1)$$

where A is the area of the enclosed optical path, Ω is the rotation rate of the two beam interferometer, λ_0 and c_0 are wavelength and light velocity in vacuum. Sagnac effect in matter is more delicate to explain, but it is completely independent of the indices of refraction or of the guidance condition, and keeps the same value as that in the vacuum.²⁾ For a monochromatic wave divided into two beams by a beam splitter placing in a two-beam-interferometer, the typical response is;

$$I_D = K(1 + \cos \phi), \quad (2)$$

where K , peak photocurrent, is the function of both spectral responsivity of photodiode, and the amplitude of interfering monochromatic beams, ϕ is the phase difference between two monochromatic beams. The generated photocurrent $I_D(t)$ in an interferometer to which reciprocal phase modulation is applied;

$$I_D(t) = I_0[1 + \cos(\phi_R + \phi_{m0} \sin \omega_m t)], \quad (3)$$

where $\phi_{m0} \sin \omega_m t$ is the periodic phase bias created by the sine wave biasing modulation applied to phase modulator such as piezoelectric transducer (PZT) or electrooptic modulator (such as LiNbO_3).³⁾ The sine wave modulation/demodulation technique is sufficient to defeat zero response problem of the interferometer resulting from $N\pi$ Sagnac phase shift (N is an odd integer) and to clarify the direction of the rotation rate.

If eq. (3) is expanded series of n th order Bessel functions, even and odd frequency harmonics are obtained and it can be seen that the odd frequency harmonics ω_m carries the Sagnac phase shift information with sine function, making IFOG

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the direction sensitive-absolute angular rate sensor.⁶⁾ The resultant photocurrent;

$$I_D'(t) = 2KJ_1 \left(2\phi_{m0} \sin \frac{\omega_m \tau}{2} \right) \sin \omega_m t \cdot \sin \phi_R, \quad (4)$$

where $2\phi_{m0} \sin(\omega_m \tau/2)$ is the nonreciprocal phase difference, τ is the loop transit time of the sensing coil, and J_1 is the first order of Bessel function and takes the maximum value at

$$2\phi_{m0} \sin \frac{\omega_m \tau}{2} = 1.8 \text{ rad}. \quad (5)$$

When the nonreciprocal phase difference goes to zero, the interference signal begins to disappear. Therefore the intensity of the photocurrent resulting from the odd frequency harmonics, carrying Sagnac phase, can be maximized by adjusting $\omega_m = 2\pi f_m = 2\pi(1/2\tau)$.⁶⁾ After the completion of band-pass filtering and demodulation steps for the signal coming from the open-loop IFOG in turn, Sagnac phase shift carried by the odd frequency components is obtained as a function of sine. Although sine function exhibits a solution to determine the rotation direction of an open-loop IFOG the linearity of the dynamic range of the open-loop IFOG is wholly restricted the periodic behavior of the sine function " $\sin \phi_R$ "⁴⁻⁶⁾ and the linearity of the scale factor of the open-loop IFOG is significantly disturbed by large Sagnac phase shifts, induced by relatively high rotation rates.

Both the accuracy and the linearity of Scale Factor are those of the most critical performance parameters of an IFOG together with Random Walk induced by thermal,⁷⁾ photon-shot,⁸⁾ and excess noise⁹⁾ in unit of $^\circ/\text{h}/\text{Hz}^{1/2}$ or $^\circ/\text{h}^{1/2}$, and bias stability (long term drift) induced by magneto-optic Faraday effect, nonlinear Kerr effect and Shupe effect, which create lack of reciprocity^{4,6,10)} in unit of $^\circ/\text{h}$.

After demodulation of the photocurrent coming from IFOG the voltage contain optical and electrical scale factors together with optical and electrical offsets:

$$V_{\text{de mod}}(\Omega) = A_1 \sin(A_2 \Omega + A_3) + A_4 \quad \text{V}, \quad (6)$$

where A_1 , A_2 , A_3 , and A_4 is the electrical scale factor, the optical scale factor, the optical bias (or offset), and the electrical bias, respectively.¹¹⁾ Sine function in eq. (6) causes an inherent nonlinearity in scale factor of an open-loop IFOG and the nonlinearity of scale factor not only directly affects to determine the rotation rate information but also the total angular displacement, calculated through the electronic integration (including digital signal processing methods) of the demodulation circuit output with precise clock generation. Every past error resulting from this nonlinearity on the dynamic range of an open-loop IFOG causes the errors in the determination of the total angular displacement because of time and voltage integration.⁶⁾

The constraint related to the nonlinearity of the dynamic range of an open-loop IFOG is overcome by closed-loop IFOG approach.

2. Description of ADCL-IFOG Prototype

For ADCL-IFOG prototype, a superluminescent light

emitting diode (SLED), manufactured by Superlum Inc., and coded as SLD-561-MP2-DIL-SM, is used as low coherent optical source the driving current and cooling currents of which are controlled by 89C51 microcontroller. The SLED package has metal outer surface and in order to get good temperature stabilization and so optical intensity and wavelength stabilization, besides microcontroller controlled thermoelectric cooler current, SLED is located in an aluminium heat sink, which isolates the SLED package from the environmental temperature changes. The averaged optical power incident on the pigtailed InGaAs in Fig. 1 is approximately $62 \mu\text{W}$, centered at 1288.9 nm , the spectral bandwidth of which is 34.6 nm . Therefore the temporal coherence length of the source is $48 \mu\text{m}$ and by means of this low coherent source an interference pattern is observed with high contrast.

PZT is engaged as phase modulator to form sine wave biasing modulation on IFOG prototypes. Several turn of sensing coil optical fiber is wrapped on PZT to attain adequate modulation depth in interference pattern.

The cut-off wavelength of the sensing coil optical fiber is 1250 nm so the single mode transmission is assured for the SLED emission. This wavelength and the spectral bandwidth do not disturb the reciprocity. Sensing coil consists of 1700 m long single mode optical fiber wrapped on a mandrel the diameter of which is 15 cm . Even though the laboratory, where IFOG prototype are constructed, is a temperature controlled-environment, the sensing coil is of dipolar winding, to impede the thermal induced phase shifts in sensing coil and to compensate the temperature transient effects.

In the configuration to assure ideal number of cross/bar passing of counter propagating waves, two 3-dB couplers are used and a fiber optic polarizer is placed between them so as to remove/minimize circular polarized components which creates strong polarization influence on the bias stability and the scale factor accuracy.

The minimum response time of the sensing coil against the applied rotation rate is $8.5 \mu\text{s}$, the loop transit time. 3-dB frequency of the low pass filter, used for suppressing the high frequency component $\sin^2 \omega_m$ and minimizing the fluctuations at the output of the demodulation circuit with AD630, is set to 10 krad/s , which corresponds to 1.592 kHz . The frequency of sine wave bias modulation applied to PZT is $f_m = 65.411 \text{ kHz}$ for 1700 meter long sensing coil, therefore this low pass filter suppresses f_m components with an attenuation of -32.3 dB .

2.1 Principle of ADCL-IFOG prototype operation

The principle of closed-loop approach is to make rotation induced-Sagnac phase shift zero by manipulating the phase modulator with a feedback voltage. Rotation rate Ω affects Sagnac phase shift ϕ_R and of course the interference pattern, which is caused by sine wave modulation, as a linear change together with an optical offset, eqs. (1) and (6). The linear relationship between Sagnac phase shift and rotation rate enables to manipulate Sagnac phase shift linearly by a suitable feedback operation and it is therefore possible to defeat inherent nonlinearity of scale factor of an open-loop IFOG, with help of a closed-loop approach.

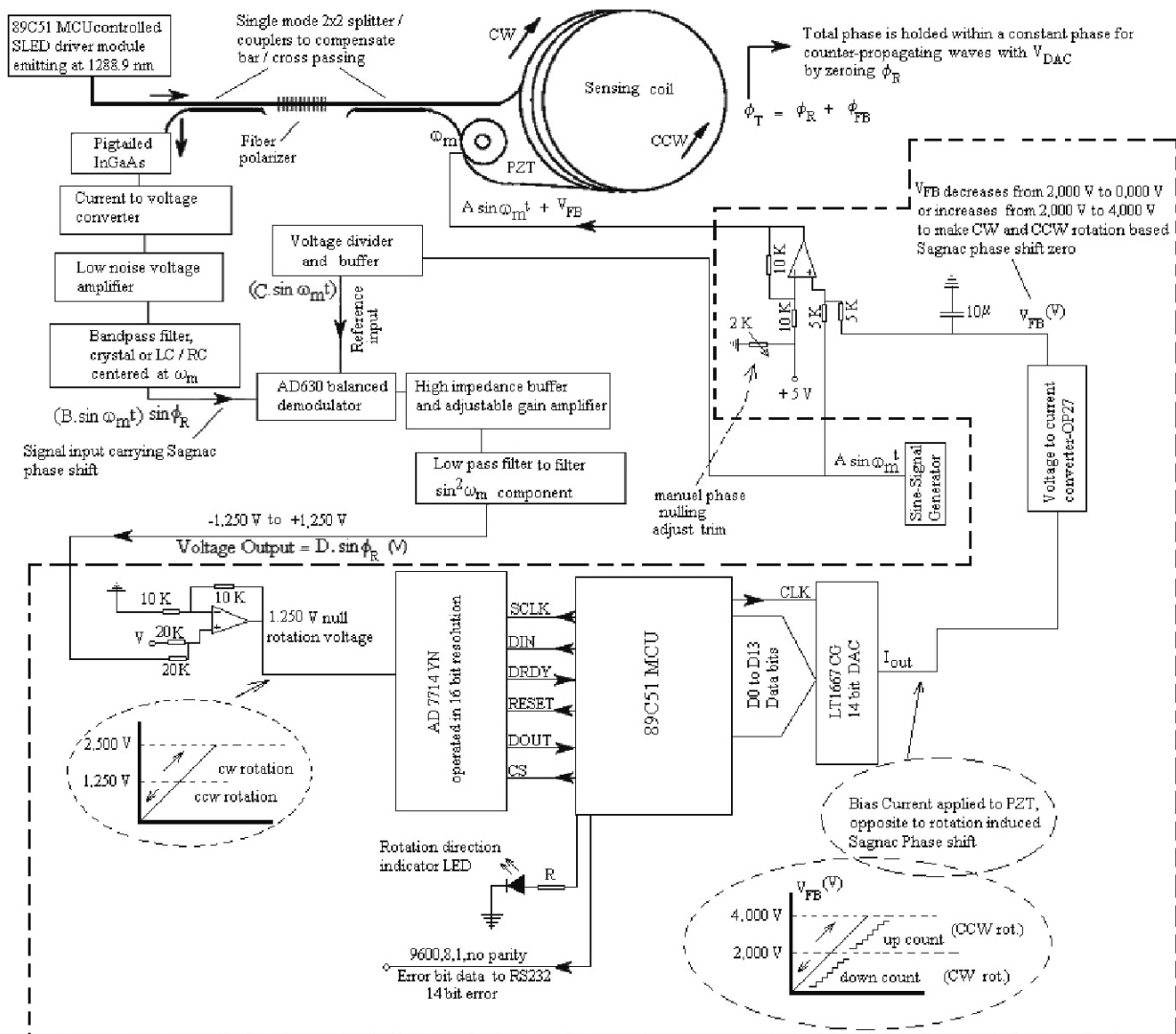


Fig. 1. ADCL-IFOG prototype. Dashed border line shows the feedback circuit, controlled by 89C51 microcontroller.

Even though this approach is necessary for obtaining more linear scale factor, it is not individually sufficient to get more accurate scale factor results. Taking into account the components of eq. (6), besides electrical drifts in the demodulation circuit, the optical source-based effects like wavelength and intensity stability, and the optogeometrical variations on the sensing coil caused by the nonlinear Faraday, Kerr and Shupe effects result in the changes on both accuracy and linearity of scale factor of an IFOG regardless of closed-loop or open-loop.¹²⁾

The output of the demodulation circuit is adjusted to ± 1.25000 V by adjustable gain amplifier. To hold the output of the demodulation circuit at positive when IFOG is subjected to clockwise (CW) and counter clockwise (CCW) rotations, this output is fed into a linear summing circuit with a reference voltage of $+1.25000$ V. Therefore the null rotation rate voltage of the IFOG is $+1.25000$ V and the full range takes places 0.00000 to 2.50000 V.

The voltage output is sampled by AD7714YN 24-bit sigma-delta analog to digital converter (ADC), which is

operated at 16-bit mode, and it can be easily to be achieved by writing “Filter High Register” of AD7714YN, which has a series of on-chip serial registers. As compared with the other type ADCs (flash, multistage, successive-approximation register) the sigma-delta ADCs, which have higher resolution through their digital filtering, have relatively lower sampling speeds. In this application the reference voltage is 2.50000 V and 2.4576 MHz crystal is used for the clock generation. For a full scale of 2.50000 V, the step size of AD7714YN equals to 38 μ V. In order to run AD7714YN, the necessary pins to be connected to microcontroller (or bus/port of a PC) were introduced in Fig. 1 RESET, CS (Chip Select), DRDY (Data Ready) inputs are active LOW signals. DIN (Data In) and DOUT (Data Out) pins run fully synchronously with rising edge of SCLK signal. When DIN pin is used to write the configuring data into the on-chip registers of AD7714YN the data bits proportional to input voltage are read from DOUT pin after DRDY pin goes to LOW state. The LOW state of DRDY output of AD7714YN demonstrates that the data bits are ready for reading. For this

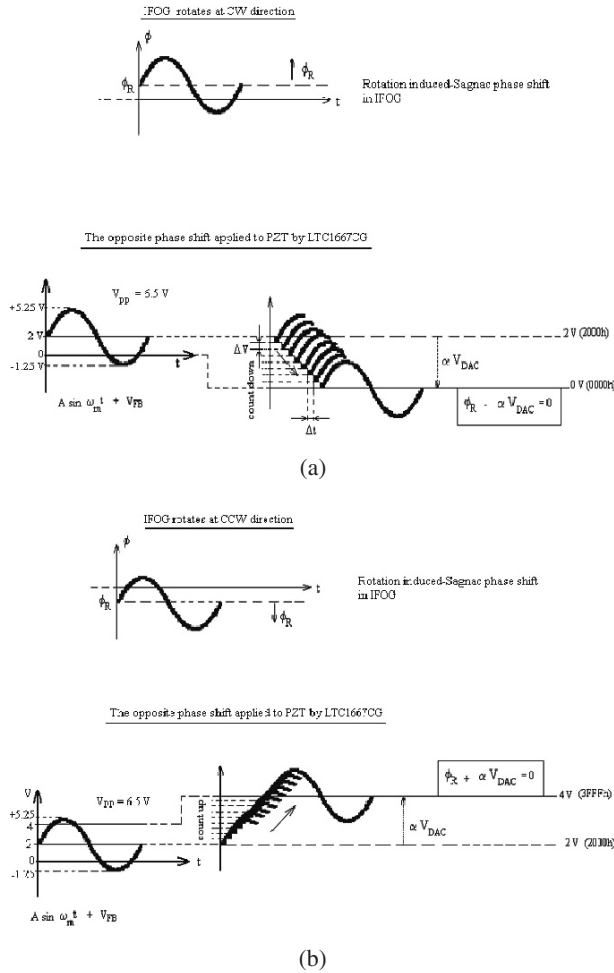


Fig. 2. (a) The Sagnac Phase Nulling voltage applied to the phase modulator of the ADCL-IFOG prototype rotating at CW direction. (b) The Sagnac phase nulling voltage applied to the phase modulator of the ADCL-IFOG prototype rotating at CCW direction.

application 2.50000 V at the input of AD7714 corresponds to $(11111111111111)_2 = (\text{FFFFh}) = 2.50000 \text{ V}$. The null rotation voltage is 1.25000 V and the binary data which corresponds to the null rotation voltage is $(1000000000000000)_2 = (8000\text{h}) = 1.25000 \text{ V}$. The duration of DRDY high state is $500 \times t_{\text{clk in}} = 205 \mu\text{s}$ as in ref. 13. In this application, a write operation to AD7714YN and a read operation from it take a 120 and $30 \mu\text{s}$, respectively. Total time duration consumed by AD7714 is about $360 \mu\text{s}$ for one sampling.

89C51 microcontroller which is different from that of SLED, and has a special software designed to control both 16-bit AD7714 ADC and LTC1667CG 14-bit DAC synchronously. The microcomputer holds $(8000\text{h}) = 1.250000 \text{ V}$, corresponding to the null rotation rate and loads 14-bits $(1000000000000000)_2 = (2000\text{h}) = 2.0000 \text{ V}$ to directly the input pins of LTC1667, which generates an output current proportional to its input data bits, and supplies with a dc bias voltage for PZT with current to voltage converter.¹⁴⁾ During operation of IFOG the microcomputer

compares continuously the voltage data carrying Sagnac phase shift, sampled by AD7714, with the 16-bit null rotation binary data $(8000\text{h}) = 1.250000 \text{ V}$ in real time. With this comparison the microcomputer determines which direction the IFOG rotates and in order to null the rotation induced-Sagnac phase shift, a counting which creates the opposite phase shift on PZT, and which is to be loaded on the input pins of LTC1667CG, is started by the microcomputer, beginning from (2000h) . This counting is downward from $(2000\text{h}) = 2.0000 \text{ V}$ to $(0000\text{h}) = 0.0000 \text{ V}$ for zeroing the Sagnac Phase in CW direction and is upward $(2000\text{h}) = 2.0000 \text{ V}$ to $(3FFF\text{h}) = 4.0000 \text{ V}$ for zeroing the Sagnac phase shift in CCW direction. The total phase in IFOG ϕ_T ;

$$\phi_T = \phi_R + \phi_{FB} \quad \text{deg}, \quad (7)$$

where ϕ_{FB} is the feedback phase, created by LTC1667 output.⁶⁾ Therefore ϕ_T is written in place of ϕ_R in eq. (3) and the total phase ϕ_T is induced to zero or an offset by the feedback phase, resulting in Sagnac phase shift ϕ_R to zero, for closed-loop approach.

$$\phi_T = \phi_R + \alpha(2.0000 + V_{\text{DAC}}) \quad \text{deg}, \quad (8)$$

where $(2.0000 + V_{\text{DAC}})$ is V_{FB} in Fig. 1 and the total phase shift on phase modulator can be written as in eq. (8) in terms of voltage. Where α is a constant which relates the voltage to the phase and so, LTC1667CG adds $\alpha \cdot V_{\text{DAC}}$ to the total phase through PZT. When the IFOG is stationary, the AD630 demodulation circuit output is 1.25000 V and the Sagnac Phase ϕ_R is zero. Due to no counting, V_{DAC} is zero: In this case, there is a constant phase of $\alpha \cdot 2.0000$ ($^\circ$) in the IFOG as an offset phase for counter-propagating waves only.

$$\phi_T = \alpha \cdot 2.0000 \quad \text{deg}, \quad (9)$$

when ADCL-IFOG rotates in CW direction [Fig. 2(a)], AD7714YN sends the binary data higher than $(8000\text{h}) = 1.250000 \text{ V}$, yielded by demodulation circuit, to microcontroller. The microcontroller decides the direction of rotation by comparing the new data stream with the data previously loaded (8000h) and then counts down by beginning from (2000h) , corresponding to 2.0000 V , and send the each count to LTC1667CG by synchronizing with CLK signal to compensate the rotation induced-Sagnac phase shift. The microcontroller checks the output of AD7714YN whether or not 1.250000 V. If not, the counting down is maintained. In this case the spontaneous phase change in phase modulator is written as follows;

$$\phi_T = \phi_R^{\text{CW}} - \alpha V_{\text{DAC}} + \alpha \cdot 2.0000 \quad \text{deg}. \quad (10)$$

As soon as the output of AD7714YN is $1.250000 \text{ V} = (8000\text{h})$ the counting down is completed by microcontroller, the error voltage bits are sent to serial buffer register of microcontroller, and the total phase remains constant " $\alpha \cdot 2.0000$ ($^\circ$)". When the IFOG is kept stationary, microcontroller output and so, LTC1667CG input go back (2000h) , resulting in 2.0000 V at the output of current to voltage converter in the feedback circuit, default value.

In case of rotation of ADCL-IFOG on CCW direction [Fig. 2(b)] AD7714YN sends the binary data lower than (8000h) = 1.250000 V, yielded by demodulation circuit, to microcontroller. The microcontroller decide the direction of rotation by comparing the new data stream with the data previously loaded (8000h) and then counts up by beginning from (2000h), corresponding to 2.0000 V and send the each count to LTC1667CG by synchronizing with CLK signal, to compensate the rotation induced-Sagnac phase shift. In this case the spontaneous phase change in phase modulator is written as follows;

$$\phi_T = \phi_R^{CCW} + \alpha V_{DAC} + \alpha 2.0000 \text{ deg.} \quad (11)$$

As soon as the output of AD7714YN is 1.250000 V = (8000h) the counting up is completed by microcontroller, the error voltage bits are sent to serial buffer register of microcontroller, and the total phase remains constant “ $\alpha 2.0000$ (°)” again. When IFOG is kept stationary, microcontroller output and so, LTC1667CG input go back (2000h), resulting in 2.0000 V at the output of current to voltage converter in the feedback circuit, default value. The each voltage value deviating from 2.0000 V, which occurs on both CW and CCW rotation direction, is an error voltage, which is defined as V_{DAC} . Considering eqs. (10) and (11), eq. (12) can be written as follows due to the fact that LTC1667CG produces the same change in step size for one bit change. α (°/mV), which is a constant parameter relying on the optical, electrical and geometrical characteristics of the ADCL-IFOG, relates the error voltage V_{DAC} to the rotation induced Sagnac phase shift.

$$\phi_R^{CW} - \alpha V_{DAC} = \phi_R^{CCW} + \alpha V_{DAC}, \quad (12)$$

$$\phi_R^{CW} = -\phi_R^{CCW} = \phi_R \text{ deg,} \quad (13)$$

$$\alpha = \frac{\phi_R(\Omega)}{V_{DAC}} \text{ deg/mV.} \quad (14)$$

The scale factor (SF) of the ADCL-IFOG prototype is defined as the ratio of the applied rotation rate to the error voltage produced by LTC1667CG;

$$SF = \Omega/V_{DAC} \text{ (°/h)/mV.} \quad (15)$$

The full range bias to the phase modulator is 4.0000 V, LTC1667CG forms 16383 (14 bits resolution) steps and every step equals to 244 μ V.

The time constant (R.C) of the low-pass filter, is within 100 μ s. This value is important to determine whether or not the dc output of the demodulation circuit reaches to the sufficient level at the output of the low pass filter. To achieve it, the microcontroller is held at NOP (no operation) for 610 μ s just after the phase step is applied to PZT so as to the dc output of demodulation circuit attains to the required voltage level due to the time constant of the low pass filter. Taking into account that the total duration of write/read operation and sampling cycles of AD7714YN is 360 μ s, it is vital to synchronize LTC1667CG with the sensing coil, demodulation circuit and AD7714YN. It is achieved by adjusting 1 machine cycle of 89C51 microcontroller to 1 μ s (more realistic value is 1.0850694 μ s) by using Special Function Registers

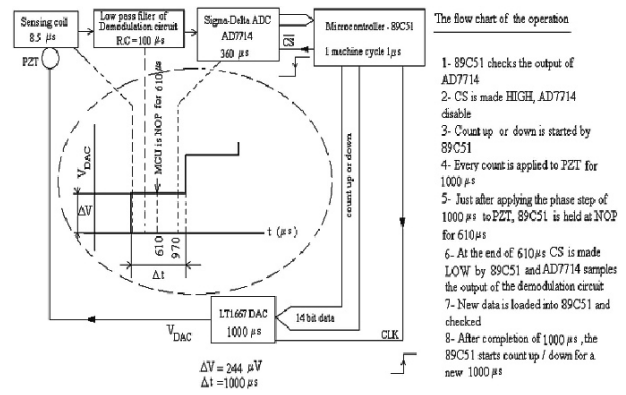


Fig. 3. Timing and flow chart of operation for the ADCL-IFOG prototype.

(TMOD, TCON, TH0, and TL0 registers) of 89C51 with 11.0592 MHz clock.

3. Results

In order to reduce Kerr effect, two 2×2 couplers used in IFOG were so selected that the optical power difference between two arms of 2×2 coupler is the smallest at the end of a series of optical power measurement at 1288.9 nm. Therefore the reciprocity arising from refractive index change, Kerr effect-based, was minimized.

Regarding magneooptic Faraday effect, the single mode sensing coil was covered by μ -metal (Cu) and it was settled in a Faraday cage. Therefore the effect of earth magnetic field change on the portion of the circular polarization fields propagating in the single mode sensing coil was minimized.

To relieve Shupe effect (variation longitudinally and cross sectionally and refractive index change in the sensing coil, depending on temperature and acoustic effects), the sensing coil was wrapped in a manner of symmetric dipolar winding so the sensing coil was symmetrically subjected to temperature and acoustic variations for CW and CCW waves.

In order to minimize noise causing random walk, it is necessary to make successive and suitable operations in optical side and electronic side.

In optical side, to reduce photon shot noise of SLED package was first located inside of a Al heat sink so as to isolate it from environmental temperature variations. Then the temperature information inside the SLED package was fed back by built-in negative temperature coefficient (NTC) thermistor into 89C51 microcontroller. Microcontroller supplies cooler current to built-in thermoelectric cooler (TEC) through 8-bit DAC and darlington transistor couple according to the output of a fast comparator circuit, reference pin of which was set to a reference resistance, corresponding to 31 °C. Therefore by keeping the interior temperature of SLED package at (31.0 ± 0.5) °C, the photon density in a unit of time was remained relatively stable. Noise Equivalent Rotation rate $NE\Omega$ at photon shot noise threshold for the proposed ADCL-IFOG;

$$NE\Omega = \frac{\Omega_{\min}}{\sqrt{B}} = \frac{\lambda c_0}{8\pi AN} \sqrt{\frac{2h\nu}{P_0\eta}} \quad (^\circ/\text{h})/\text{Hz}^{1/2} \quad (16)$$

$\lambda = 1288.9 \text{ nm}$, $A = 0.01766 \text{ m}^2$ (radius of sensing coil 7.5 cm), $N = 1062$ turn wraps, $c_0 = 3.10^8 \text{ m/s}$, $h = 6.62 \times 10^{-34} \text{ J}\cdot\text{s}$, $\nu = 2.3275 \times 10^{15} \text{ Hz}$, InGaAs photodiode quantum efficiency $\eta \approx 1$, and optical power falling onto InGaAs photodiode $P_0 = 62 \mu\text{W}$. In this case the ultimate rotation rate which can be sensed at the threshold of photon shot noise is theoretical $NE\Omega = 6.59 \times 10^{-4} (^\circ/\text{h})/\text{Hz}^{1/2}$ within a bandwidth of sampling duration of 1 s ($B = 1$).

Excess Noise is reversely proportional to the spectral bandwidth of the optical source and it decreases with the increasing spectral bandwidth. Considering that increasing the spectral bandwidth of low coherent source restricts IFOG dynamic range ($^\circ/\text{h}\text{--mV}$), and thermal noise (Detector thermal noise–Johnson noise) is dominant over excess noise for the optical power levels under $5 \mu\text{W}$, for the proposed ADCL-IFOG an optimization was made by means of a series of bias stability measurements. As a result, for a spectral bandwidth of 34.5 nm , optical power of $62 \mu\text{W}$ was applied to the proposed ADCL-IFOG.

In addition, free arms of the two 2×2 couplers were tightly wrapped in order to impede Fresnel reflections at the air/glass interfaces and spurious standing waves. Moreover, low coherence length of optical source inherently reduces the backscattering inside the sensing coil, guided to the InGaAs photodiode.

In electronic side, the demodulation circuit, which provides narrow noise bandwidth by means of the bandpass filter inherently,¹⁵⁾ was well-grounded and powered by highly stabilized and filtered $\pm 15 \text{ V}$ dc source. The operational amplifiers (opamps) used in demodulation circuit are OPA111s and OP27s, which have low bias currents and offset voltages. All the passive circuit elements are precision elements and so exhibits low temperature dependence. In order to evaluate the effect of Boltzmann (kT) induced-the electrical noise, the output of feedback circuit output (DAC) was sampled via $8(1/2)$ digit voltmeter when there is no optical power applied in the gyro system. The output deviates maximum 0.150 mV from zero rotation level, corresponding to a rotation rate of $1.13 (^\circ/\text{h})$ (4 V full scale and 14bit resolution DAC) regarded as electrical noise contribution.

Photodetector, pigtailed InGaAs, was not supplied with a reversed DC bias so that shot noise, based on dark current, is not induced. Therefore detector noise figure is only limited with kT based-Johnson noise. Noise equivalent power (NEP) of the pigtailed InGaAs is an order of $10^{-14} \text{ W/Hz}^{1/2}$ (small active area). The temperature stability of the location, where the measurements, were carried out is $(23.0 \pm 1.0) ^\circ\text{C}$.

Additional information about sensing coil, demodulation circuit (dynamic phase tracking circuit) and practical application of IFOG, depending on the bias stability, was given in ref. 15.

3.1 Bias stability measurement result of ADCL-IFOG

Before bias stability measurement of ADCL-IFOG given

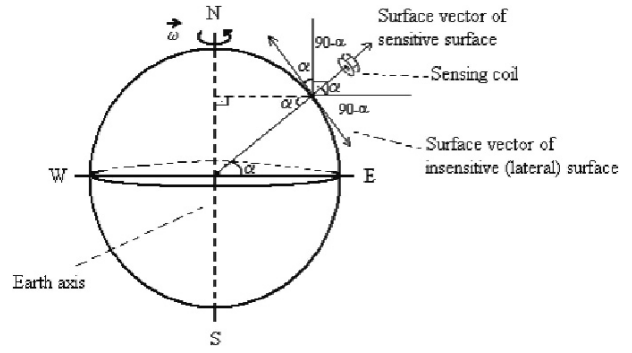


Fig. 4. Settling of the ADCL-IFOG before measuring the bias stability and determining the scale factor to avoid earth rotation rate induced contribution.

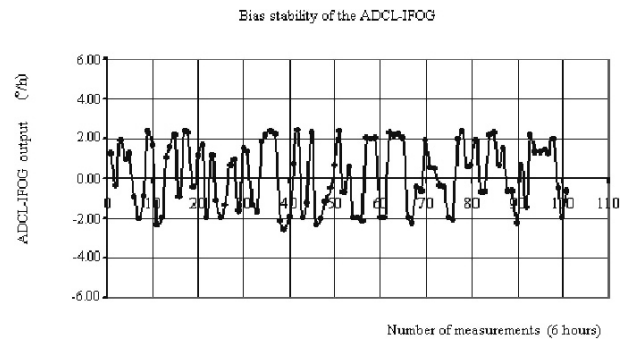


Fig. 5. Bias stability of the ADCL-IFOG prototype at $23.0 ^\circ\text{C}$.

in Fig. 1, it is so positioned that the scalar product of $\mathbf{A}$ and $\mathbf{\Omega}_{\text{earth}}$ in eq. (1) equals to zero. The latitude, where the measurements are carried out, is 40.8° for Gebze/KOCAELİ (Fig. 4).

According to this configuration the sensing coil of the prototype is tilted 40.8° from the northwards. After completion of 3 h conditioning of ADCL-IFOG, the bias stability measurements are carried out. In the measurements, the error voltage (V_{DAC}) is sampled and collected by $8(1/2)$ digit voltmeter for a period of 6 h .

100 samplings, each of which is within a bandwidth of 1 Hz , constitute every voltage data, corresponding to bias rotation. The bias stability of the prototype is calculated with Allan Variance method;

$$A\text{VAR}^2(t) = \frac{1}{2(n-1)} \sum_i [V_{i+1}(t) - V_i(t)]^2, \quad (17)$$

where $A\text{VAR}^2$ is Allan Variance as a function of time t , V_i is the i th average value of the error voltage V_{DAC} and n is the total number of data collected.¹⁾ The bias stability of the ADCL-IFOG was calculated as $1.48 (^\circ/\text{h})$ by using the data which form the bias stability graphic given in Fig. 5.

3.2 Scale factor measurement result of ADCL-IFOG prototype

In order to derive the scale factor of the ADCL-IFOG in terms of Sagnac Phase Nulling voltage V_{DAC} (error voltage)

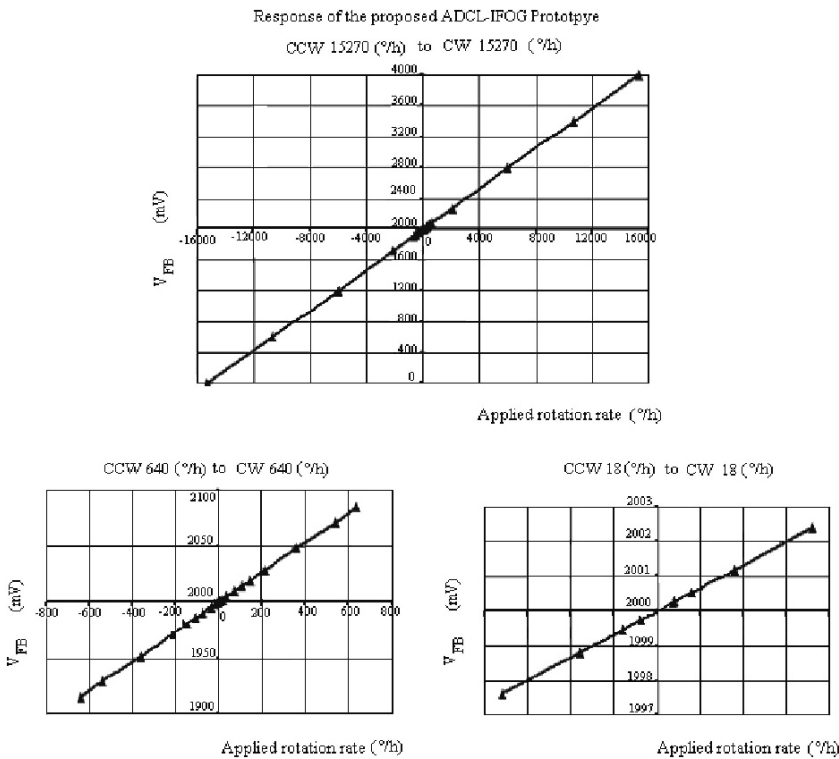


Fig. 6. The Scale Factor of the ADCL-IFOG prototype vs the applied rotation rate.

applied by LTC1667 connected to 89C51, it is subjected to the stable rotation with varying from 1 to 15270 (°/h) by keeping the position which eliminate the earth rotation rate contribution. During the stage of the scale factor measurements of the ADCL-IFOG, the rotation rate applied by servo motor plate having an accuracy of 0.02 (°/h) is remained stable until the binary data, which made Sagnac Phase zero, is generated by microcontroller. The voltage outputs of the feedback circuit versus the stable rotation rate applied were presented on Fig. 6.

Considering 1000 μ s phase step of LTC1667CG, 16383 steps of 14 bit resolution takes 16.4 s approximately. That is, the full voltage range to be applied to PZT is 0.0000 to 4.0000 V and this range is completed 16.4 s. The rotation rate of 15270 (°/h) equals to ~ 4.2 (°/s) and therefore an angular displacement larger than 4.2 (°/s) $\times$ 16.4 s = 68.9 (°) is allowed to nullify Sagnac phase shift for the ADCL-IFOG during the count up/down of microcontroller. Averaged scale factor of ADCL-IFOG is derived as 7.53888 (°/h)/mV versus the applied rotation rate with a relative standard deviation of 0.85% (Table 1).

4. Conclusion

In this paper, the details belonging to an ADCL-IFOG were presented together with bias stability (long term stability) measurement and the determination of scale factor of the ADCL-IFOG prototype.

In order to avoid inherently nonlinear behavior of a conventional open-loop IFOG, which especially take places in larger Sagnac phase shifts induced by higher rotation rates, an ADCL-IFOG with a digital feedback circuit, composed of AD7714YN (operated in 16-bit resolution)

and LTC1667CG (14 bit DAC) synchronously controlled by 89C51 microcontroller, was designed for use in the metrological purposes such as angle measurements together with precise frequency generator. For a Sagnac phase shift range of 0.00115 to 17.57448 (°) in CW and CCW rotation directions, instead of an increasing trend of scale factor expected in a conventional open-loop IFOG which results from the demodulation of interferometer signal, to obtain ADCL-IFOG scale factor having a relative standard deviation of 0.85% showed that it had high linearity in dynamic range with a Sagnac phase shift of 35.41 (°). With relative standard deviation of 0.85%, the averaged sensitivity of the ADCL-IFOG in unit of voltage (error voltage) was calculated as 132.65 μ V/(°/h), which equals to a scale factor (SF) of 7.53888 (°/h)/mV.

For the ADCL-IFOG, minimum rotation rate of 1.90 (°/h) was nullified by applying the feedback voltage, which corresponds to the opposite phase shift to the rotation induced Sagnac phase shift, to PZT through the digital feedback circuit with LTC1667CG and a bias stability of 1.48 (°/h) was reached with a maximum peak to peak noise of 4.97 (°/h) at 23.0 °C. In order to evaluate the temperature dependent-performance of the ADCL-IFOG in terms of bias stability and peak to peak noise, the temperature of the location, where the ADCL-IFOG was settled, was varied among 23.0–42.0 °C. Even though relative increase became in noise of ADCL-IFOG, peak to peak noise and bias stability didn't exceed 6.66 and 1.69 (°/h) respectively, thanks to symmetric dipolar winding of the sensing coil.

Considering the earth rotation rate of 15.041 (°/h) according to WGS84 (World Geodesic System 1984), the

Table 1. The Sagnac Phase Nulling Voltages of the proposed ADCL-IFOG prototype vs the rotation rate.

Rot. Direction	Ω (°/h)	ϕ_R (°)	V_{FB} (mV)	V_{DAC} (mV)	SF Ω/V_{DAC} (°/h)/mV	α (°/mV)
CW	15270	17.57448	2.2184	1997.7816	7.643478	0.00879699
	10650	12.25725	602.3453	1397.6547	7.619908	0.00876987
	6000	6.90549	1196.5376	803.4624	7.467680	0.00859467
	2100	2.41692	1723.2291	276.7709	7.587502	0.00873257
	640	0.73659	1914.7961	85.2039	7.511396	0.00864498
	540	0.62149	1928.9820	71.0180	7.603704	0.00875122
	360	0.41433	1951.7853	48.2147	7.466606	0.00859343
	216	0.24860	1971.5194	28.4806	7.584108	0.00872867
	144	0.16573	1980.6413	19.3587	7.438513	0.00856110
	108	0.12430	1985.5970	14.4030	7.498417	0.00863004
	72	0.08287	1990.5385	9.4615	7.609758	0.00875819
	36	0.04143	1995.1826	4.8174	7.472885	0.00860066
	18	0.02072	1997.6215	2.3785	7.567846	0.00870995
	9	0.01036	1998.7980	1.2020	7.487521	0.00861750
	4	0.00460	1999.4726	0.5274	7.584149	0.00872871
	2	0.00230	1999.7363	0.2637	7.584108	0.00872867
	1.84	0.00212	1999.7542	0.2458	7.485639	0.00533776
CCW	1.84	0.00212	2000.2471	0.2471	7.447679	0.00531070
	2	0.00230	2000.2684	0.2684	7.452372	0.00857705
	4	0.00460	2000.5287	0.5287	7.565128	0.00870682
	9	0.01036	2001.1927	1.1927	7.546099	0.00868492
	18	0.02072	2002.4013	2.4013	7.495820	0.00862705
	36	0.04143	2004.8075	4.8075	7.488350	0.00861846
	72	0.08287	2009.5354	9.5354	7.550842	0.00869038
	108	0.12430	2014.3840	14.3840	7.508327	0.00864145
	144	0.16573	2018.9787	18.9787	7.587450	0.00873251
	216	0.24860	2028.5681	28.5681	7.560891	0.00870195
	360	0.41433	2048.1947	48.1947	7.469705	0.00859700
	540	0.62149	2071.3202	71.3202	7.571493	0.00871415
	640	0.73659	2085.2387	85.2387	7.508327	0.00864145
	2100	2.41692	2275.1709	275.1709	7.631620	0.00878335
	6000	6.90549	2805.2006	805.2006	7.451559	0.00857611
	10650	12.25725	3395.8547	1395.8547	7.629734	0.00878118
	15270	17.57448	3998.3820	1998.3820	7.641182	0.00867654

bias stability value, scale factor linearity, minimum rotation rate, and the temperature dependence the ADCL-IFOG prototype can be proposed for an absolute angular displacement sensor in metrology and other related fields.

One of the largest advantages of the proposed ADCL-IFOG over the analog ramp phase technique, which has an analog demodulator generating phase error signal, is to direct AD7714YN and LTC1667CG by the software in EEPROM of 89C51 microcontroller. With this direction to be performed by means of serial I/O of 89C51, digital filtering codes on chip serial registers of AD7714YN can be changed remotely. Therefore noise reduction can be directly performed with changing of digital filtering codes of AD7714YN and the dynamic range of the whole system can easily be varied within a limited range. Another advantage is to store the voltage generated by the demodulation circuit fed to AD7714YN on internal RAM of 89C51 as binary codes and is to use them to calculate the average voltage data for every rotation rate. This digital integration

and averaging of voltage data corresponding to Sagnac phase shift contribute to increase the Signal to Noise Ratio of the whole system. Hence the averaging process, easily changing of digital filtering codes of AD7714YN, changing of sampling and averaging time of AD7714YN and storage of voltage data series by software direction through serial input/output (I/O) of 89C51 gain inherently advantages the proposed ADCL-IFOG over the analog systems. Moreover to change driving and cooling currents fed to SLED package together with time durations, which binary codes match and is controlled by a second 89C51, by updated software through serial I/O without disturbing the hardware configuration of the ADCL-IFOG is an additionally practical advantage.

References

- 1) D. Loukianov, R. Rodloff, H. Sorg, and B. Steiler: *Optical Gyros and Their Applications* (The Research and Technology Organization (RTO) of NATO, Canada Communication

- Group, 1999).
- 2) W. K. Burns: *Optical Fiber Rotation Sensing* (Academic Press, San Diego, 1994).
 - 3) K. T. V. Grattan and B. T. Meggitt: *Optical Fiber Sensor Technology* (Chapman&Hall, London, 1995).
 - 4) B. Culshaw: Meas. Sci. Technol. **17** (2006) R1.
 - 5) B. Y. Kim and H. J. Shaw: Opt. Lett. **9** (1984) 263.
 - 6) H. Lefevre: *The Fiber-Optic Gyroscope* (Artech House, Boston, London, 1993).
 - 7) Hamamatsu Photonics Infrared Detectors Product Catalogue, Japan (1997).
 - 8) R. A. Bergh, H. C. Lefevre, and H. J. Shaw: J. Lightwave Technol. **2** (1984) 91.
 - 9) K. Iwatsuki: IEEE Photonics Technol. Lett. **3** (1991) 281.
 - 10) V. Dangui, H. K. Kim, M. J. F. Digonnet, and G. S. Kino: Opt. Express **13** (2005) 6669.
 - 11) R. P. Moeller, W. K. Burns, and N. J. Frigo: J. Lightwave Technol. **7** (1989) 262.
 - 12) H. K. Kim, J. F. Digonnet, and G. S. Kino: J. Lightwave Technol. **24** (2006) 3169.
 - 13) AD 7714 product data sheet of Analog Devices Inc. (2008).
 - 14) LTC1667 product data sheet of Linear Technology Inc. (2008).
 - 15) O. Celikel: Opt. Quantum Electron. **39** (2007) 147.